SQL Fundamentals for Data Analysts

This guide explains core SQL concepts in simple language, provides practical examples, and demonstrates how the concepts work together in real-world data analysis.

1. What is SQL?

SQL (Structured Query Language) is the language used to communicate with databases.

Why use SQL?

- Retrieve data
- Insert data
- Update data
- Delete data
- Analyze data

Example:

```
SELECT * FROM Customers;
```

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2. Difference Between WHERE and HAVING

WHERE filters rows before aggregation.

Example:

```
SELECT *
FROM Employee
WHERE Salary > 50000;
```

HAVING filters groups after aggregation.

Example:

```
SELECT Department,
    AVG(Salary) AS Avg_Salary
FROM Employee
GROUP BY Department
HAVING AVG(Salary) > 50000;
```

Rule:
WHERE = Row filter
HAVING = Group filter

3. Primary Key

A Primary Key uniquely identifies each row in a table.

Rules:

- Cannot be NULL
- Cannot repeat
- One primary key per table

Example:

```
CREATE TABLE Student(
   Student_ID INT PRIMARY KEY,
   Name VARCHAR(50)
);
```

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4. Foreign Key

A Foreign Key creates a relationship between tables.

Example:

```
CREATE TABLE Orders(
   Order_ID INT PRIMARY KEY,
   Student_ID INT,
   FOREIGN KEY(Student_ID)
   REFERENCES Student(Student_ID)
);
```

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5. Composite Key

A Composite Key uses two or more columns together to uniquely identify a record.

Example:

```
PRIMARY KEY(Student_ID, Course_ID)
```

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6. Surrogate Key

A Surrogate Key is a system-generated key.

Examples:

IDENTITY

AUTO_INCREMENT

SEQUENCE

Example:
Customer_ID = 1,2,3,4...

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7. GROUP BY

GROUP BY combines rows that have the same value.

Example:

```
SELECT Department,
    SUM(Sales) AS Total_Sales
FROM Sales_Table
GROUP BY Department;
```

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8. BETWEEN

BETWEEN retrieves values within a range.

Example:

```
SELECT *
FROM Employee
WHERE Salary BETWEEN 50000 AND 100000;
```

Equivalent to:

```
Salary >= 50000
AND Salary <= 100000
```

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9. ORDER BY

ORDER BY sorts data.

Ascending:

```
SELECT * FROM Employee
ORDER BY Salary ASC;
```

Descending:

```
SELECT * FROM Employee
ORDER BY Salary DESC;
```

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10. IN

IN checks if a value exists in a list.

Example:

```
SELECT *
FROM Employee
WHERE Department IN ('IT','HR','Finance');
```

Sample Employee Dataset

Emp_ID | Name | Department | Salary | Age
1 | John | IT | 70000 | 28
2 | Mary | HR | 55000 | 35
3 | James | IT | 90000 | 40
4 | David | Finance | 45000 | 30

Complete Script Using Multiple Concepts Together

```
SELECT Department,
      COUNT(*) AS Total_Employees,
      AVG(Salary) AS Avg_Salary
FROM Employee
WHERE Age BETWEEN 25 AND 45
  AND Department IN ('IT','HR','Finance')
GROUP BY Department
HAVING AVG(Salary) > 50000
ORDER BY Avg_Salary DESC;
```

How SQL Executes This Query

Step 1: FROM Employee
Step 2: WHERE Age BETWEEN 25 AND 45
Step 3: GROUP BY Department
Step 4: Calculate COUNT and AVG
Step 5: HAVING AVG(Salary) > 50000
Step 6: SELECT columns
Step 7: ORDER BY Avg_Salary DESC

Practice Questions

1. Find employees earning above 60,000.
2. Find employees aged between 25 and 40.
3. Count employees by department.
4. Show departments with average salary above 70,000.
5. Sort employees by salary from highest to lowest.